

Feature Extraction

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May 2017

Abstract

Feature extraction is a procedure aimed at selecting and transforming a data set in order to increase the performance of a pattern recognition or machine learning system. Nowadays, since the amount of data available and its dimension is growing exponentially, it is a fundamental procedure to avoid overfitting and the curse of dimensionality, while, in some cases, allowing a interpretative analysis of the data. The topic itself is a thriving discipline of study, and it is difficult to address every single feature extraction algorithm. Therefore, we provide an overview of the topic, introducing widely used techniques, while at the same time presenting some domain-specific feature extraction algorithms. Finally, as a case, study, we will illustrate the vastness of the field by analysing the usage and impact of feature extraction in neuroimaging.

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I. INTRODUCTION

IN machine learning and pattern recognition, feature extraction is the procedure through which the performance of a predictive model is improved via constructing an optimal set of variables. A predictive model, or predictor, can be a regressor, classifier or any machine that can be used to extract information from known data and predict the output of new, unseen data. As for the term feature, it refers to all variables or attributes that are collected so that our predictive model can identify a certain sample or subject. For example, in a medical environment, our sample would be a patient that visits a doctor -our predictive model-, and its features would be all symptoms that, either told or acquired by the doctor, help him making a diagnosis.

Formally, let us note our set of N samples as a $K \times N$ matrix $\mathbf{X}$, being the n -th sample a column vector $\mathbf{x}_n = \{x_0^n, x_1^n, \dots, x_K^n\}^T$ containing K features. Therefore, the feature extraction problem deals with the construction of an optimal subset $\mathbf{S}$ (size $C \times N$), where each sample $\mathbf{s}_n = \{s_0^n, s_1^n, \dots, s_C^n\}$ is redefined in terms of a new set of features.

Historically, data was manually collected and filtered, leading to a optimized small number of features that could be easily analysed. Since then, more and more data has become available, and currently it is common to find a scenario where tens of thousands and even millions of features are available. This is the case of image processing, where images contain millions of pixels, or email classification algorithms where words are features. This might theoretically increase the accuracy of our predictors, but in practice poses a major problem: the curse of dimensionality [1].

The curse of dimensionality refers to a problem that occurs when the ratio sample size to feature size is small. In contrast to *dense spaces*, where a set of N lower-dimensional samples are close to one another, the distances between samples in higher-dimensional spaces are huge, leading to what is known as *almost empty spaces*.

Traditionally, classifiers did not generalize well in these almost empty spaces, leading to poor performance and a high generalization error. To overcome the problem, it might be necessary to ‘fill’ the feature space with new samples, in order to obtain a more dense space, and therefore a classifier with lower generalization error. Newer approaches included reducing the number of features in order to provide a lower-dimensional representation of the original set [2].

Today, due to the amount of data available, feature extraction is mostly used for feature reduction. However, there are still cases where new features, build from the data, are used along the original data in

order to increase the performance of our predictor. Attending to the type of algorithms used to perform feature extraction, we could distinguish two major groups: decomposition methods, that decompose the data as a combination of hidden, unknown features, and data-driven methods, which includes a wide variety of methods that take advantage of the data type that is being analysed. Some examples of the first are Principal Component Analysis (PCA) or Linear Discriminant Analysis (LDA) [3], whereas some examples of the latter are geometric shape identifiers in image processing [4] or Cepstral analysis in speech recognition [5].

In this work, we will start with an overview of the main categories of feature selection methods. Later, in Section III we will introduce some widely used decomposition algorithms, and finally, in Section IV, we will address more application-specific methodology.

II. FEATURE SELECTION

Feature selection is the first strategy used for feature reduction, and it is often used along with feature extraction in order to build more complex pattern recognition systems. It refers to any strategy intended to find a subset of the original features containing the more suitable ones according to a certain criterion. Therefore, irrelevant features are discarded, and resultant models are faster and more cost-effective [6]. However, it usually requires an additional optimization to find the parameters for the optimal feature subset, and furthermore, it is impossible to guarantee that the optimal features for the subset are the same of the full feature set [7].

Feature selection algorithms are often classified into three categories [8], depending on how the algorithm is integrated in the classification model: filtering, wrappers and embedded approaches. See Figure 1 for a visual description of these approaches.

II.1. Filtering

Filtering methods are based on the computation of a feature relevance score directly on the data, before any interaction with the classification model. The relevance score is used to sort the different features, discarding those with a lower score, and it is usually computed independently for each feature, in what is called a univariate approach [8]. Common univariate filtering methods include Correlation, Mutual Information [9], Euclidean Distance [10], or any Hypothesis testing strategy [11] such as χ^2 , t -Test, Fisher's Discriminant Ratio (FDR) or others.

Multivariate approaches that look at the interactions between different features are being developed as well, although they are slower and less scalable than the aforementioned techniques. They include a wide range of techniques, e.g. Analysis of Variance (ANOVA) [12] or correlation-based approaches [13].

II.2. Wrappers

Similarly to filtering methods, wrappers are a feature selection techniques where a certain score is assigned to each feature. This score is used to choose those features that are more useful in our model, but in contrast to filtering methods, the score used here is generally an estimation of the performance of the features in the chosen predictive model [14]. Performance is usually assessed via cross-validation.

A very simple wrapper is, for example, a search guided by accuracy, where the score used to select features is the accuracy obtained in a per-feature classification. Some other commonly used techniques are greedy search algorithms: Forward Selection (FS) and Backward Elimination (BE) [6]. FS evaluates the performance of a classifier using one feature and sequentially adding new ones, in order to obtain a optimal set of features that maximizes the performance of the model. On the contrary, BE starts with the full set of features and sequentially removes features while maintaining or increasing the performance.

Apart from FS and BE, other widely known wrappers include genetic algorithms [14], category utility [15] and the expectation-maximization algorithm [16].

II.3. Embedded approaches

Embedded approaches try to combine the advantages of both filters and wrappers, at the same time that they interact with the proposed model. In embedded approaches, features are added or removed from the feature subset during the building and evaluation of the model. While similar to wrappers, embedded approaches perform feature selection while building the predictive model, whereas wrappers use the space of all possible feature subsets. This makes a more efficient use of the data and allow the model to reach an optimal subset faster.

Some popular embedded approaches are Random Forest [17], a classifier comprising many decision trees, where the best decision tree defines the optimal feature subset, or using the SVM weight vectors to select the feature subset [18].

III. DECOMPOSITION METHODS

Decomposition methods are the most common feature extraction technique. They intend to model a dataset as a combination of several unknown, common variables, which allows the representation of the dataset in a new space defined by K variables.

Mathematically, the decomposition process could be generally defined as:

$$\mathbf{x}_n = t_n^0 \mathbf{q}_0 + t_n^1 \mathbf{a}_1 + \dots + t_n^K \mathbf{a}_K + \epsilon = \mathbf{t}_n \mathbf{A} + \epsilon \quad (1)$$

where $\mathbf{A}$ is the inverse of the mixing matrix $\mathbf{W}$, also known as component loading matrix, $\mathbf{t}_n = \{t_n^0, t_n^1, \dots, t_n^K\}$ is the representation of the n -th sample in the transformed space defined by the new features, also known

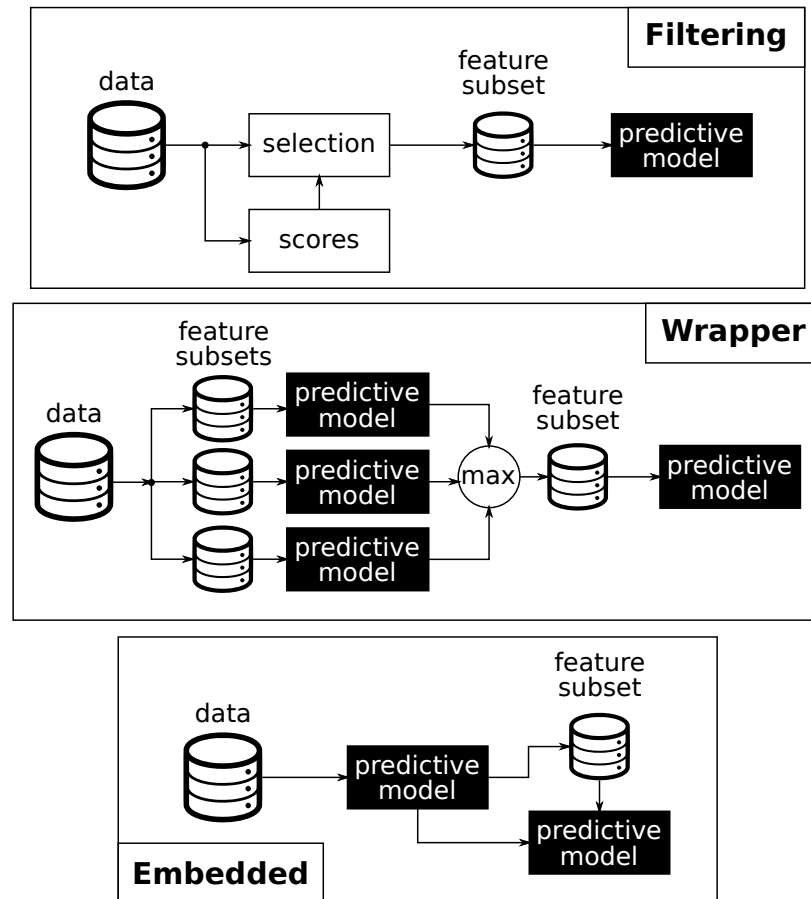


Figure 1: Illustration of the three feature selection approaches.

	Advantages	Disadvantages
Filters	• Simpler to use • Fast • The optimal subset is always the same	• Less powerful
Wrappers	• Very powerful feature set (all possible sets are examined) • The optimal subset is always the same	• Much slower (extensive training required)
Embedded	• Speed/power trade-off • Faster than wrappers	• Optimal set is not unique, depends on initialization

Table 1: Summary of the advantages and disadvantages of the methods proposed in Sections II.1, II.2 and II.3.

as component score matrix and ϵ is an estimation of the reconstruction error. The representation of the n -th sample $\mathbf{x}_n$ in the transformed space would be:

$$\mathbf{t}_n = \mathbf{x}_n \mathbf{W} \quad (2)$$

In this work, we will describe two widely used approaches in more detail, and afterwards, we will provide an introduction to many other decomposition techniques used in the literature. The two approaches will be Principal Component Analysis (PCA), a unsupervised technique, and Linear Discriminant Analysis (LDA), a supervised case.

Supervised learning stands for any learning method that includes known information (e.g. classes) to optimize the model. In our case, unsupervised decomposition techniques blindly extract features only using the information contained in the data, whereas supervised techniques obtain class-optimized features. In general, supervised techniques are used for classification, whereas unsupervised approaches have a wider range of applications, including exploratory data analysis or blind source separation.

III.1. Principal Component Analysis

Principal Component Analysis (PCA) is probably the most widespread decomposition method. It is widely used in image processing [4, 19], medicine [3], gene analysis [20] and many other applications. PCA is an unsupervised, that is, it performs the decomposition without any further knowledge about the samples in contrast to supervised methods, where information such as class is accounted to optimize the decomposition.

PCA is intended to decompose the dataset as a linear combination of principal component loadings, chosen so that they maximize the variance explained. Thus, each subject's original p -dimensional feature vector $\mathbf{x}_n$ could in theory be reconstructed using a series of coefficients $\mathbf{s}_n$ (namely principal component scores). In practice, however, a truncated version of PCA is often used, where only the first K components are used ($K < p$), and the truncated mixing matrix $\mathbf{W}$ (of size $p \times K$) contains the transformation of the p original features to the new K -dimensional space.

Mathematically, the mixing matrix $\mathbf{W}$ is the set of eigenvectors obtained after applying the eigenvalue decomposition to the covariance matrix of the data $\mathbf{X}^T \mathbf{X}$ (assuming that the column mean was subtracted from the data) so that:

$$\mathbf{X}^T \mathbf{X} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^T \quad (3)$$

where $\mathbf{\Lambda}$ is a diagonal matrix whose diagonal elements contain the eigenvalues of each principal component, and the mixing matrix $\mathbf{W}$ contains the eigenvectors.

The most popular approach to obtaining the decomposition takes advantage of some analogies between the Singular Value Decomposition (SVD) of $\mathbf{X}$ and the Eigenvalue Decomposition of $\mathbf{X}^T \mathbf{X}$. Since the eigenvalue decomposition is often more computationally expensive, the SVD approach is often used. SVD performs the decomposition:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \quad (4)$$

where $\mathbf{U}$ and $\mathbf{V}$, the left and right singular matrices, are square matrices containing n and p singular vectors (of size n and p respectively) whereas $\mathbf{\Sigma}$ is a rectangular n by p diagonal matrix containing the singular values of $\mathbf{X}$. Using that decomposition, the covariance matrix

$$\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{\Sigma} \mathbf{U}^T \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T = \mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^T \quad (5)$$

which, comparing with Equation 3, shows that we can obtain the mixing matrix as $\mathbf{W} = \mathbf{V}$, the eigenvalues $\mathbf{\Lambda} = \mathbf{\Sigma}^2$, and the transformed samples $\mathbf{T} = \mathbf{X} \mathbf{W} = \mathbf{U} \mathbf{\Sigma}$.

One particular field where PCA has been continuously and successfully applied is face recognition [21, 22]. In this approach, a common set of so-called *eigenfaces* [22] is generated from the training dataset, and each face is represented as a shorter vector containing the component scores of each eigenface. These eigenfaces are the eigenvectors (contained in $\mathbf{W}$), rearranged so as to form a two dimensional image again. An illustration of this problem using the Olivetti dataset [21] can be found in Figure 2.

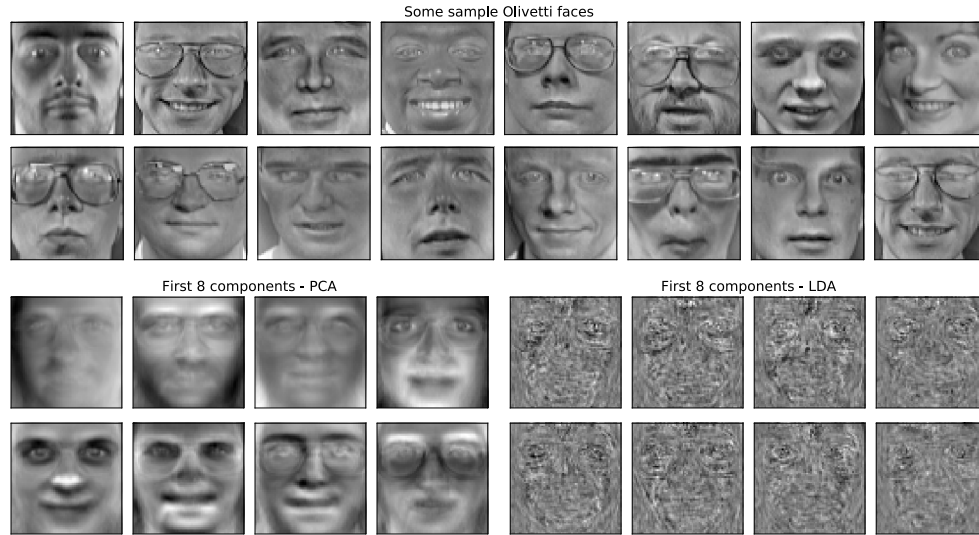


Figure 2: An overview of the six first faces from the Olivetti dataset [21], the six first eigenfaces resulting of applying PCA, and the six first discriminant when applying LDA to differentiate between subjects (credit: AT&T Laboratories Cambridge).

III.2. Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a technique similar to PCA, but supervised. It tries to find the perfect mixing matrix that emphasizes the class separation in the data, instead of maximizing the variance captured by each principal component [23]. The objective is that our linear discriminant matrix $\mathbf{W}$ maximizes the between-class distance and minimizes the within-class distance.

LDA assumes normally distributed data and statistically independent features, similarly to Quadratic Discriminant Analysis (QDA). However, it makes an additional homoscedascity assumption, that is, that the class covariances are identical and have full rank [23], yielding a linear discriminant function.

To perform LDA for C classes, first the scatter between-class $\mathbf{S}_b$ and within-class $\mathbf{S}_w$ matrices (representing the between-class and within class-distance respectively, both of size $p \times p$) are computed. To do so, we must represent the samples in data/class pairs as $(\mathbf{x}_n, y_n)$, where $y_n \in \{1, 2, \dots, C\}$ is the class of the n -th sample. That way, the matrices are defined as:

$$\mathbf{S}_b = \sum_{c=1}^C \sum_{n: y_n=c} (\mathbf{x}_n - \boldsymbol{\mu}_c)(\mathbf{x}_n - \boldsymbol{\mu}_c)^T \quad (6)$$

$$\mathbf{S}_w = \sum_{c=1}^C N_c (\boldsymbol{\mu}_c - \boldsymbol{\mu})(\boldsymbol{\mu}_c - \boldsymbol{\mu})^T \quad (7)$$

where $\boldsymbol{\mu}$ is the mean vector, $\boldsymbol{\mu}_c$ is the mean vector of all samples belonging to class c and N_c is the number of samples belonging to class c .

Afterwards, an eigenvalue decomposition of the product $\mathbf{S}_w^{-1}\mathbf{S}_b$ is performed, and the eigenvectors are

sorted by decreasing eigenvalues. Then, the matrix $\mathbf{W}$ is constructed by choosing the K first eigenvectors with largest eigenvalues, yielding a matrix of size $p \times K$, just like in PCA. The transformed dataset would therefore be obtained in the same way, using $\mathbf{T} = \mathbf{XW}$.

Given their similarities, people often ask whether PCA is better than LDA or otherwise. It depends much on the application, and the need of a supervised or unsupervised feature extraction method. It was shown that PCA outperforms LDA when the number of samples per class is small [24]. However, they are often used together [25, 3, 26] in many applications, using PCA for a preliminar feature reduction and then LDA to enhance class separation. In Figure 2, the eigenfaces obtained using LDA are compared to those using PCA.

III.3. Other Decomposition Methods

As previously, there are two major approaches to decomposition methods: supervised and unsupervised. The use of either one or the other depends fundamentally on the application, and sometimes is more a philosophical topic.

PCA is the most prominent unsupervised approach, but there exist many other widely used decomposition algorithms, especially Factor Analysis (FA) [27, 28] and Independent Component Analysis (ICA) [29, 11]. Factor Analysis has often been considered analogous to PCA, since they both model the contribution of some hidden variables to the variance of the observed data. In FA, the observed data is decomposed as a linear combination of Factors (similar to principal components), however the algorithm accounts for the random noise inherent in every measurement, which affects the computation of the variance explained by each factor. However, it requires to formulate a good initial model, since it is necessary to perform assumptions about the covariance matrix. On the contrary, PCA does not make any assumptions about this at all. In general, it is suggested that Factor Analysis works best when there is a solid theoretical model, whereas PCA is to be used as a general feature extraction technique and to explore patterns in the data [30].

Independent Component Analysis performs another component decomposition of data that, due to its properties, was widely used specially in the early 2000s [29, 11]. ICA assumes non-gaussianity and provides statistically independent components, unlike PCA where the components were only uncorrelated. The algorithm usually involves centring and whitening the data (subtracting the mean and standardizing the data), and is very frequent to apply dimensionality reduction via SVD or PCA. The cost function used to assess independence depends on the algorithm, but InfoMax and Negentropy are the most extended [31].

PCA, LDA, FA and ICA model the data as a linear combination of hidden variables, but sometimes its linear nature has been considered insufficient to explore more complex data. In addition to QDA (introduced in Sec. III.2), a number of extensions to PCA have been introduced to account for nonlinearities in the data. One usual approach is kernel PCA, that allows to deal with nonlinearities via a kernel within the PCA algorithm itself [3].

There exist many other algorithms that perform a decomposition of $\mathbf{X}$, for example Non-Negative Matrix Factorization (NNMF) [32], used where the non-negativity of the data and the resulting decomposition is guaranteed; or Partial Least Squares (PLS) [33], a supervised approach that defines one model for data and other for predictions, in what is known as bilinear model, and is increasingly popular.

IV. DOMAIN-SPECIFIC FEATURE EXTRACTION

Depending on the domain and type of data used, more specific approaches can be made. In machine learning, it is common to process 1-dimensional (audio, seismographic), 2-dimensional (images, microarrays), 3-dimensional (video, medical imaging) or even 4-dimensional signals (functional neuroimaging, 4D ecography, etc). In this case, the traditional approach would be to consider all pixels (or voxels, for volumetric imaging) as individual features and input them to a decomposition methods. However, this removes significant information, for example their timing or location, that are inherent to each modality.

Alternatively, more specific feature extraction algorithms that take advantage of these particularities can be defined, providing a meaningful extraction of features that could improve the performance of our predictive model.

IV.1. Audio Processing

Audio, speech and, in general, one-dimensional signal processing was perhaps the first domain where feature extraction algorithms were used [34, 35, 36, 37]. Many algorithms have been used to deal with problems such as automatic speech recognition [34, 36, 38, 39, 40, 41, 42], speaker identification [37] or music categorization [35, 43, 44].

Attending to the transformation that they perform on the original signal, there is a major group that extracts features based on the frequency and spectrum of the signal.

IV.1.1 Spectral Features

Spectral features have one thing in common: they are based on the Fourier transform of the signal, either in its continuous or discrete version. The Fourier transform allows to decompose a signal in its spectral components; the logarithm of the squared magnitude of the Fourier transform is known as the power spectrum.

This power spectrum has many properties, the main of which is to identify the frequency components that allows us to model things such as timbre or pitch. These coefficients have been used in voice and music processing [40]. A useful tool that represent the evolution of the power spectrum through time is the spectrogram (see Figure 3).

Going a little further, we define the Cepstrum, the inverse Fourier transform of the power spectrum. This is a useful tool to characterize the evolution of pitch along time [36]. Based on Cepstrum and using

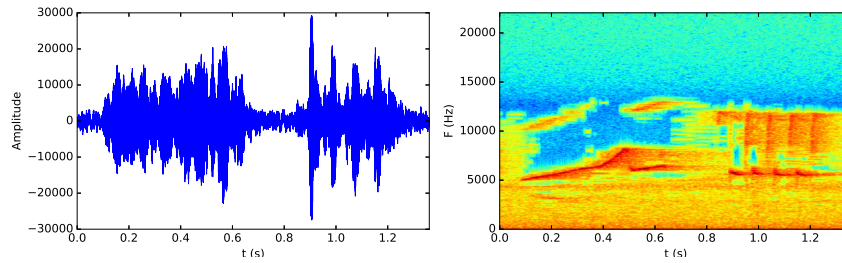


Figure 3: Example of the spectrogram of an audio signal.

the Mel-Frequency (a non-linear perceptual scale of frequency), other features are extracted using the Mel-Frequency Cepstral Coefficient (MFCC), used to encode and recognize voice [42] and measure the timbre of music [43].

Voice signals are often represented by means of Linear Predictive Coding (LPC), also known as autoregressive (AR) modeling [38]. In this approach, the voice spectrum is modelled as the output of an all-pole filter, whose coefficients can be used as features.

IV.1.2 Other Audio Features

Other audio feature extraction range from low level features such as energy or pitch to complex and abstract features such as the results of applying deep neural networks. The simplest approaches include computing the correlation [36], rhythm, zero-crossing frequency, or by detecting voice activity [39]. The coefficients of the Wavelet transform have also been used in a similar way to the Fourier transform [35].

IV.2. Image Processing

In image processing and computer vision, the data is presented in bidimensional (or higher-dimensional) arrays where each element in a given location represent a pixel intensity. Histograms and colour histograms are an widespread tool, since they represent the frequency of different intensity values (see Figure 4). They have been heavily used in image processing and classification [45], although they entail spatial information loss.

Pixel location is usually ignored when the data arrays are vectorized, for example, in the computation of a histogram. Other image specific approaches usually take advantage of this information and incorporate it to extract features that model shape or texture.

IV.2.1 Texture Features

Many texture feature extraction algorithms exist, including wavelets or fractals, but among them, the statistical estimators based on the Gray Level Co-occurrence Matrix (GLCM) are the most extended [46, 47, 48]. In short, the GLCM quantifies the frequency of transitions between the intensity levels I_i and I_j in the

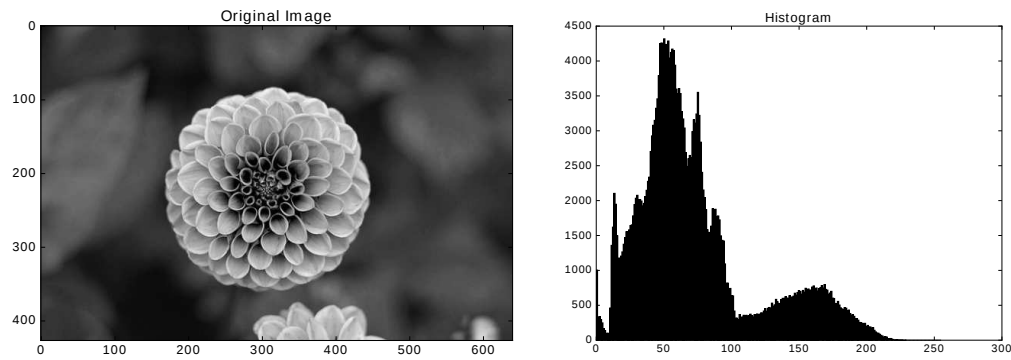


Figure 4: Example of the histogram of a greyscale image.

bidimensional data array $\mathbf{X}$ in a certain direction $\vec{\Delta}$. The element in the position (i, j) of the matrix is computed as:

$$C_{\Delta}(i, j) = \sum_{p \in \mathbf{X}} \begin{cases} 1 & \text{if } \mathbf{X}(p) = I_i \text{ and } \mathbf{X}(p + \Delta) = I_j \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

where p is a vector containing the coordinates within the image array $\mathbf{X}$. The data array in $\mathbf{X}$ is frequently quantized into 8, 16 or 32 grey levels, in order to operate with smaller matrices. Using the GLCM, we can compute further measures that characterize properties of the texture of the image, such as Entropy, Homogeneity or Contrast.

There exist other approaches to texture feature extraction, however. Among them, we find Local Binary Patterns (LBP) [49], an histogram of the distribution of intensities in the vicinity of each pixel, or fractal dimension [50, 51], an index that measures complexity by comparing the details and the scale of a certain image.

IV.2.2 Other Image Features

Finally, other features such as moments [52] (used to identify, for example, area, centre of mass and axes) have been used in different applications. Local features characterizing individual objects in segmented images are also widespread, e.g. brightness and channel standard deviation[50], shape features such as aspect ratio, circularity or area [53, 54, 50] or even measures based on differential geometry, such as torsion and curvature [55].

IV.3. Convolutional Neural Networks

Convolutional Neural Networks (CNN) are a hot topic these days [56, 57, 58, 59], since they are the basis of the new paradigm known as Deep Learning [60]. The term Deep Learning refers to those learning algorithms that build hierarchical knowledge from the data, from simple to complex patterns, usually

structured in layers. The higher the number of layers, the more complex this knowledge will be, yielding to high level abstractions and a “deeper” understanding of the data. That is where the term “Deep” comes from, and this “depth” is usually related to the number of layers used, and therefore, to the complexity examined. CNN are mainly used in image processing for classification, while providing more and more abstract features from the input images in their deeper layers.

The basis are those of Artificial Neural Networks (ANN). In a nutshell, artificial neurons take a series of inputs, add them in a weighted approach and fire an activation function yielding a output. Among the most used firing functions are tanh, the sigmoid and lately the Rectifier Linear Unity (ReLU). These neurons are grouped in layers, which are then interconnected with other layers. Traditionally, ANN used a small number of layers, since the performance did not increase when incrementing the number of layers.

CNNs, however, introduce many changes that make them more flexible as well as yielding unprecedented performance. First, there are more types of layers, among them convolutional layers, in which neurons perform the convolution between the input and a filter; pooling layers, in which the results of the convolution are downsampled and pooled together; or ReLU layers, that perform the aforementioned classification. See Figure 5 for an example of CNN architecture.

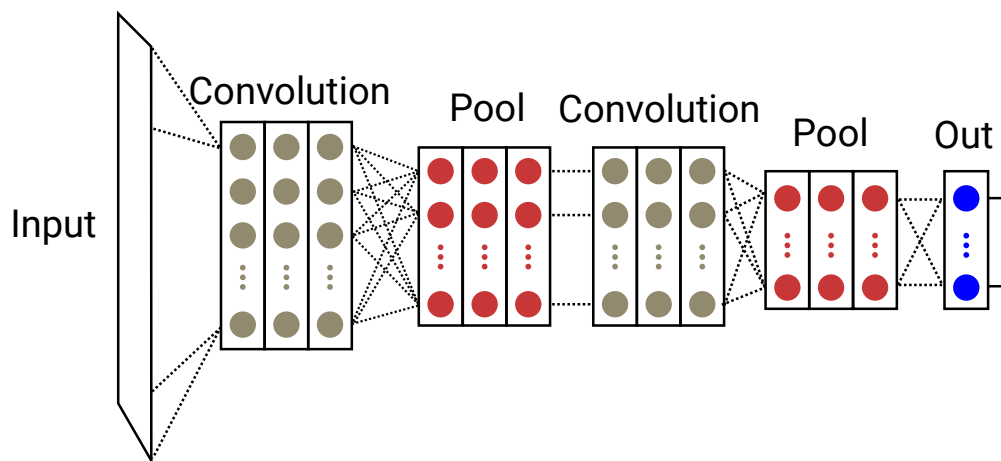


Figure 5: Example of a convolutional Network with two convolution layers (each with three filters), two pooling layers and one output (dense) layer.

Different combinations of these types of layers provide an internal representation of different features in images, that allow pattern detection with unprecedented accuracy [56, 58, 60]. Therefore, CNN act as an embedded approach where feature extraction, feature selection and classification is performed at the same time. The internal features represented in each layer have been widely used as a visual tool in many applications [56, 57, 58, 59], as they represent abstractions on the data otherwise impossible to extract.

V. CASE STUDY: FEATURE EXTRACTION IN NEUROIMAGING

Medical imaging is an interesting field where feature extraction is frequently applied. Data acquired here range from pure two-dimensional images, such as radiography, to four-dimensional arrays that present the evolution of three-dimensional volumetric images through time, such as the ones used in neuroimaging. The amount of data to be processed here ranges from several hundreds of pixels (in the two-dimensional images) to millions of voxels. Therefore, feature extraction algorithms are more than a need.

Neuroimaging pipelines are similar to any other machine learning pipeline: a subset of the input data is selected to train a classifier and then test its performance. Classifiers were the object of early improvements, using only feature selection, such as in the widely extended Voxels As Features approach [61]. Lately, more and more complex feature extraction algorithms are being used.

Image decomposition is today a significant part of the improvements made in neuroimaging research. We have examples using PCA [62, 3] (see Figure 6) and LDA [3], Factor Analysis [27, 28], ICA [63, 11] or PLS [64, 65]. Some methods involve using feature selection before [11] or after [66] the decomposition.

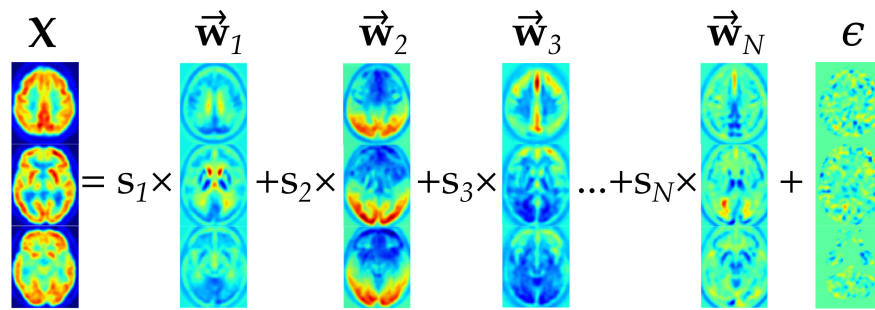


Figure 6: PCA decomposition of a functional brain image.

Inside the field, FreeSurfer is, along with Statistical Parametric Mapping (SPM), a standard piece of software. FreeSurfer [67] implements a feature extraction based on the estimation of the cortical thickness and surface area, that is, the thickness of the grey matter at each point of the brain (see Figure 7). This has helped in many studies involving neurodegeneration, for example, in the diagnosis of Alzheimer's Disease (AD) [68], Autism [69] and even in combination with decomposition algorithms like PCA [70].

Apart from the aforementioned approaches, a number of new strategies have been developed to extract new information from the samples. In [71], shape modelling was applied to the segmented Magnetic Resonance Images (MRI) to parametrize the hippocampi and effectively diagnosis AD. In [48], the authors use texture features to obtain characteristics, also applied to the diagnosis of AD. In [59], Deep Learning techniques using sparse filtering are used to extract features in Parkinson's Disease functional imaging, and in [72] a geometric approach to feature selection and transformation was developed using spherical coordinates inside the brain, creating two-dimensional feature maps from each three-dimensional neuroimage.

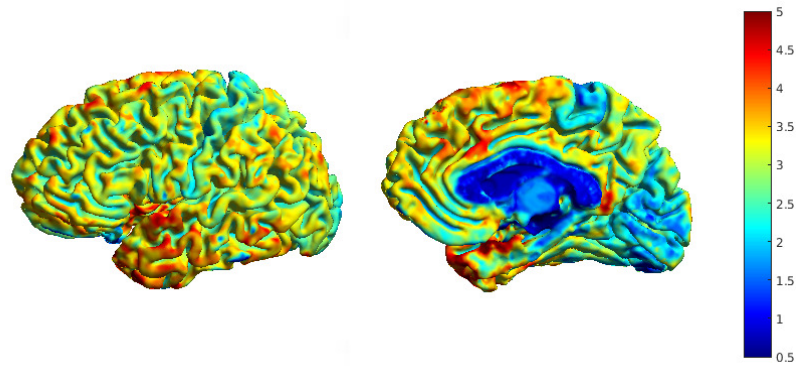


Figure 7: Cortical Thickness of a brain.

There are many more examples of feature extraction in neuroimaging. This section is intended to illustrate the vastness of feature extraction in individual applications, and how specific these solutions can be. This is the same for any other application that the reader has in mind, from face recognition to chemical identification; from gene expression to voice processing. Each field has its own particularities, and there are specific solutions to these problems.

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